

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY

COURSE CODE: CSA5115

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 2

Total Marks: 15

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I **M. Nishanth (192511154)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M Nishanth

Annexure A

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security

A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern.

- a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
- b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message

A cybersecurity intern intercepts the coded message: “GSRH RH Z HVXIVG” during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

- a. Deduce the most likely original message.
- b. Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics.

Question 1: Event Communication Security

Scenario: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, the message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern of 4 positions (Caesar Cipher).

Conceptual Background: The Caesar Cipher

The Caesar Cipher is one of the earliest known classical symmetric-key substitution ciphers, historically attributed to Julius Caesar for secure military communication. It works by replacing every letter of the plaintext with a letter found a fixed number of positions further along in the alphabet, wrapping around from Z back to A when necessary. If each letter of the alphabet is numbered from 0 (A) to 25 (Z), the encryption and decryption operations can be expressed mathematically using modular arithmetic:

Encryption: $C = (P + K) \bmod 26$

Decryption: $P = (C - K) \bmod 26$

where P is the numeric value of the plaintext letter, C is the numeric value of the resulting ciphertext letter, and K is the secret key (the shift value, here $K = 4$). Because addition and subtraction by the same key value undo one another, the Caesar Cipher is a symmetric cipher the identical key is shared secretly between sender and receiver and used for both directions of the transformation.

a. Reconstructing the Encrypted Message (Shift = 4)

Applying the encryption rule $C = (P + 4) \bmod 26$ to every letter of the plaintext "MEET AT NOON" (spaces are not part of the alphabet and are therefore left unchanged so that word boundaries remain visible to the intended recipient):

Plaintext	M	E	E	T	(sp)	A	T	(sp)	N	O	O	N
Ciphertext (+4)	Q	I	I	X	(sp)	E	X	(sp)	R	S	S	R

Worked calculation for each letter (using A=0 ... Z=25):

- $M (12) + 4 = 16 \rightarrow Q$
- $E (4) + 4 = 8 \rightarrow I$ (appears twice, giving "II")

- $T(19) + 4 = 23 \rightarrow X$
- $A(0) + 4 = 4 \rightarrow E$
- $N(13) + 4 = 17 \rightarrow R$
- $O(14) + 4 = 18 \rightarrow S$ (appears twice, giving "SS")

Final Encrypted Message: QIIX EX RSSR

b. Retrieving the Original Message

An authorized member who is aware of the encoding rule already possesses the shared secret key ($K = 4$). To retrieve the original plaintext, the member simply reverses the operation performed during encryption by subtracting the key instead of adding it, wrapping around the alphabet from A to Z where the result would otherwise become negative:

$$P = (C - 4) \bmod 26$$

Applying this to the intercepted ciphertext "QIIX EX RSSR":

Ciphertext	Q	I	I	X	(sp)	E	X	(sp)	R	S	S	R
Plaintext (-4)	M	E	E	T	(sp)	A	T	(sp)	N	O	O	N

For example, $Q(16) - 4 = 12 \rightarrow M$, $I(8) - 4 = 4 \rightarrow E$, $X(23) - 4 = 19 \rightarrow T$, and so on, which exactly restores the original sentence "MEET AT NOON". Because the transformation is fully reversible using the single shared numeric key, no separate decryption key needs to be communicated — both parties only need to have agreed on the value 4 in advance, out-of-band, before the message was ever sent.

Security Analysis and Relevance

While the Caesar Cipher is simple to implement and understand, it offers very weak security by modern standards. Because there are only 25 possible non-trivial shift values, an attacker can perform an exhaustive brute-force search (trying every shift from 1 to 25) or use frequency analysis (matching common letter frequencies in English, such as E and T, against the frequencies observed in the ciphertext) to recover the plaintext within seconds. For the student organization's use case, this is acceptable only for very low-stakes, informal secrecy among members, and would need to be replaced by a stronger algorithm (such as AES or RSA) if genuinely sensitive information needed protection from a determined adversary.

Question 2: Intercepted Suspicious Message

Scenario: A cybersecurity intern intercepts the coded message "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

Conceptual Background: The Atbash Cipher

The Atbash Cipher is one of the oldest known monoalphabetic substitution ciphers, originally used to encode the Hebrew alphabet and later adapted to Latin scripts. Its substitution rule is a straightforward mirror-reversal of the alphabet: the first letter (A) is replaced by the last letter (Z), the second letter (B) is replaced by the second-last letter (Y), and so on. Numerically, if a letter has position P (with $A = 0$ and $Z = 25$), its Atbash-substituted letter is given by:

$$C = 25 - P$$

A defining property of Atbash is that it is a reciprocal (self-inverse, or involutory) cipher: applying the exact same substitution rule a second time returns the original letter, since $25 - (25 - P) = P$. This means the same procedure is used for both encoding and decoding — there is no separate encryption/decryption key or rule to remember.

a. Deducing the Most Likely Original Message

Applying the mirror-substitution rule $C \rightarrow 25 - C$ to every letter of the intercepted ciphertext "GSRH RH Z HVXIVG":

Ciphertext	G	S	R	H	(sp)	R	H	(sp)	Z	(sp)	H	V	X	I	V	G
Plaintext	T	H	I	S	(sp)	I	S	(sp)	A	(sp)	S	E	C	R	E	T

Worked calculation for the first word, "GSRH" (using $A=0 \dots Z=25$):

- $G (6) \rightarrow 25 - 6 = 19 \rightarrow T$
- $S (18) \rightarrow 25 - 18 = 7 \rightarrow H$
- $R (17) \rightarrow 25 - 17 = 8 \rightarrow I$
- $H (7) \rightarrow 25 - 7 = 18 \rightarrow S$

This immediately produces the recognizable English word "THIS". Continuing the same substitution across the remaining groups ("RH" $\rightarrow$ "IS", "Z" $\rightarrow$ "A", and "HVXIVG" $\rightarrow$ "SECRET") yields a complete, grammatically correct English sentence.

Most Likely Original Message: THIS IS A SECRET

b. Justification of the Deduction

The following evidence, drawn from structural pattern analysis of the ciphertext, supports the conclusion that a simple monoalphabetic substitution (specifically Atbash) was used, and that "THIS IS A SECRET" is the correct recovered plaintext:

1. Preserved word structure: The ciphertext retains the exact spacing and word lengths of natural English — four groups of length 4, 2, 1, and 6 letters respectively. Classical substitution ciphers operate letter-by-letter and preserve word boundaries, unlike modern block ciphers or stream ciphers, which produce ciphertext that gives no clue about the original word lengths. This is a strong structural signal that the message was encoded with a classical technique.
2. Short-word cribs: English has an extremely limited set of common one-letter words ("A", "I") and two-letter words ("IS", "IT", "OF", "TO", "IN", etc.). The ciphertext contains a lone single-letter word "Z" and a repeated two-letter word "RH", which appears twice in the message. This repetition is a classic indicator used in cryptanalysis: the same ciphertext pattern recurring at two different points strongly suggests it maps to the same common plaintext word both times (in this case, "IS"), which is consistent with a monoalphabetic (one ciphertext letter always maps to one plaintext letter) substitution rather than a polyalphabetic or randomized scheme.
3. Repeated-letter consistency: The ciphertext letter H appears in "GSRH", "RH" and "HVXIVG"; under a monoalphabetic cipher every occurrence of H must decode to the same plaintext letter. Checking the derived mapping (H → S in every instance: "thiS", "iS", "Secret") confirms internal consistency, which would not hold if the message used a more complex, position-dependent cipher.
4. Successful mirror-alphabet test: Directly testing the reversed-alphabet (Atbash) mapping on the first ciphertext group, "GSRH", immediately produces the valid English word "THIS" rather than a random string. Since a correct guess about a cipher's rule should make the entire message resolve into meaningful text — not just an isolated word — the fact that applying the same fixed $A \leftrightarrow Z$, $B \leftrightarrow Y$, $C \leftrightarrow X$, ... rule across every remaining letter of the message also yields valid English ("IS", "A", "SECRET") confirms that Atbash, and not some other substitution, is the correct cipher.

5. Cipher characteristics matching Atbash: Atbash is a reciprocal, key-less monoalphabetic cipher historically used for quick, low-security obfuscation rather than robust confidentiality consistent with the scenario's description of a message that used only "a simple substitution mechanism rather than complex encryption." Its lack of a variable key (unlike the Caesar Cipher's shift value) also means that, once suspected, it can be verified almost instantly by direct testing, exactly as demonstrated above.

Security Analysis and Relevance

Atbash offers essentially no cryptographic security against a knowledgeable adversary: because it has no secret key at all, anyone who suspects the technique can decode the message immediately by applying the fixed mirror substitution, as the cybersecurity intern is able to do here through simple structural analysis. In a real digital forensics or network-monitoring context, this reinforces why an intercepted message that resolves so easily under classical cryptanalysis (pattern matching on word length, repeated short words, and single-letter cribs) should be treated as a low-security or informally obfuscated message rather than as strongly encrypted communication, and why modern systems rely on key-based algorithms with much larger key spaces (e.g., AES-256) to resist this kind of straightforward structural attack.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, wellstructured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation